Exhaustion Theory

A Mathematical Framework for Finite Admissible Continuation

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Abstract

We introduce *Exhaustion Theory*, a mathematical framework for processes that are *well-behaved* up to a finite horizon and then terminate, collapse, or become undefined. The primitive object of the theory is the exhaustion machine: a finite-state dynamical system whose states are partitioned into evolving states and exhausted states. Unlike the theory of limits, which studies asymptotic behaviour, and unlike singularity theory, which studies blow-up or loss of regularity, Exhaustion Theory studies finite admissible continuation. We define exhaustion machines, exhaustion time, exhaustion profiles, frontiers, ranks, morphisms, and algebraic operations. The central theorem states that a finite exhaustion machine halts on all admissible trajectories if and only if its reachable evolving subgraph is acyclic, equivalently if and only if it admits a strictly decreasing well-founded rank. Thus the Halting Problem is decidable, indeed graph-theoretic, inside the category of finite exhaustion machines. We develop min-plus and max-plus algebraic semantics for pessimistic and optimistic exhaustion, give algorithms for validation and lifetime computation, and discuss applications to resource-bounded computation, thermodynamics, artificial intelligence, machine learning, and finite-horizon control. A final section connects Exhaustion Theory with Ambiguity Theory: ambiguity concerns branch multiplicity across alternatives, while exhaustion concerns finite continuation along time. The joint theory studies ambiguous finite processes whose branches may exhaust at different horizons.

The treatise ends with "The End"

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1 Introduction

Many mathematical theories study systems that fail to continue. Limit theory studies asymptotic behaviour as a parameter tends to infinity or approaches a boundary. Singularity theory studies points at which regular descriptions break down. Computability theory studies whether a computation halts. Thermodynamics studies irreversible loss of usable work. Reliability theory studies failure times.

Exhaustion Theory isolates a different phenomenon: a process may be perfectly well-defined for finitely many steps and then simply run out. The running out may be of energy, memory, time, admissible hypotheses, computational budget, social trust, free energy, credit, liquidity, or any finite resource needed for continuation.

The central slogan is:

$$\boxed{\text{Exhaustion Theory is the mathematics of finite admissible continuation.}}$$

The finite-state-machine formulation is primary. We distinguish two kinds of states:

$$\text{evolving states} \qquad \text{and} \qquad \text{exhausted states.}$$

An evolving state has admissible future dynamics. An exhausted state has no endogenous productive continuation. This gives the primitive skeleton:

$$\boxed{\text{evolving} \longrightarrow \text{exhausted.}}$$

The theory is designed as a companion to Ambiguity Theory. Ambiguity Theory treats unresolved multiplicity as a first-class object; Exhaustion Theory treats finite continuation as a first-class object. Their joint setting studies systems that may be ambiguous across branches and exhausting along time.

2 Preliminaries

2.1 Finite State Machines

A deterministic finite state machine consists of a finite set of states Q, an input alphabet Σ, an initial state $q_0 \in Q$, and a transition function

$$\delta : Q \times \Sigma \to Q.$$

In Exhaustion Theory the transition function is generally partial:

$$\delta : Q \times \Sigma \rightharpoonup Q.$$

Partiality means that some attempted continuations are not admissible.

2.2 Directed Graphs

Every partial finite-state machine determines a directed labelled graph

$$G = (Q, T),$$

where

$$(q, q') \in T$$

if and only if there exists $a \in \Sigma$ such that

$$\delta(q, a) = q'.$$

Reachability, cycles, paths, and strongly connected components will be understood in this graph-theoretic sense.

2.3 Well-Founded Orders

A strict order $<$ on a set W is well-founded if there exists no infinite descending chain

$$w_0 > w_1 > w_2 > \cdots .$$

The natural numbers $\mathbb{N}$, with the usual order, are well-founded. So are finite products $\mathbb{N}^k$ with lexicographic order.

3 Exhaustion Machines

3.1 Definition

Definition 3.1 (Exhaustion machine)**.** An exhaustion machine is a tuple

$$\mathcal{E} = (Q, \Sigma, \delta, q_0, Q^{\text{ev}}, Q^{\text{ex}}),$$

where:

1. Q is a finite set of states;
2. Σ is a finite input alphabet or action set;
3. $\delta : Q^{\text{ev}} \times \Sigma \rightharpoonup Q$ is a partial transition map;
4. $q_0 \in Q$ is the initial state;
5. $Q^{\text{ev}} \subseteq Q$ is the set of evolving states;
6. $Q^{\text{ex}} \subseteq Q$ is the set of exhausted states;
7. $Q = Q^{\text{ev}} \sqcup Q^{\text{ex}}$.

The defining partition is:

$$Q = Q^{\text{ev}} \sqcup Q^{\text{ex}}.$$

States in Q^{ev} have possible endogenous evolution. States in Q^{ex} represent terminal, collapsed, depleted, or undefined continuation.

Definition 3.2 (Strict terminal semantics)**.** An exhaustion machine has strict terminal semantics if

$$q \in Q^{\text{ex}} \quad \Longrightarrow \quad \delta(q, a) \text{ is undefined for every } a \in \Sigma.$$

Definition 3.3 (Absorbing terminal semantics)**.** An exhaustion machine has absorbing terminal semantics if the transition map is extended to all of Q by

$$q \in Q^{\text{ex}} \quad \Longrightarrow \quad \delta(q, a) = q \quad \text{for every } a \in \Sigma.$$

Strict terminal semantics is mathematically cleaner. Absorbing terminal semantics is often more convenient for simulations.

3.2 Runs and Exhaustion Time

Definition 3.4 (Admissible run)**.** Let

$$w = a_0 a_1 a_2 \cdots$$

be a finite or infinite word over Σ. A run of $\mathcal{E}$ on w is a sequence

$$q_0, q_1, q_2, \ldots$$

such that

$$q_{t+1} = \delta(q_t, a_t)$$

whenever $q_t \in Q^{\mathrm{ev}}$ and $\delta(q_t, a_t)$ is defined.

Definition 3.5 (Exhaustion time)**.** The exhaustion time of a run is

$$\tau = \inf\{t \geq 0 : q_t \in Q^{\mathrm{ex}} \text{ or no admissible continuation exists at } q_t\}.$$

If no such t exists, we set $\tau = \infty$.

Definition 3.6 (Universal and existential exhaustion)**.** An exhaustion machine is universally exhausting from q_0 if every admissible run from q_0 has finite exhaustion time. It is existentially exhausting from q_0 if at least one admissible run from q_0 has finite exhaustion time.

The universal notion is the structural analogue of guaranteed termination. The existential notion is the analogue of possible termination.

4 Axioms of Exhaustion Theory

Axiom 1 (State partition)**.** Every state is exactly one of evolving or exhausted:

$$Q = Q^{\mathrm{ev}} \sqcup Q^{\mathrm{ex}}.$$

Axiom 2 (Exhausted non-evolution)**.** An exhausted state has no endogenous productive continuation:

$$q \in Q^{\mathrm{ex}} \implies \mathrm{Future}(q) = \varnothing$$

under strict semantics, or only the self-loop $q \mapsto q$ under absorbing semantics.

Axiom 3 (Finite admissibility)**.** The evolving region is finite:

$$|Q^{\mathrm{ev}}| < \infty.$$

Axiom 4 (Well-founded descent)**.** For a genuine exhaustion process, there exists a well-founded set W and a rank map

$$R : Q^{\mathrm{ev}} \to W$$

such that every evolving transition strictly decreases rank:

$$q \to q', \quad q, q' \in Q^{\mathrm{ev}} \implies R(q') < R(q).$$

Axiom 5 (Loss of continuation)**.** At exhaustion, the future transition structure is lost:

$$q \in Q^{\mathrm{ex}} \implies \mathrm{Future}(q) = \varnothing.$$

Axiom 6 (Compositionality)**.** Sequential, choice, and parallel combinations of exhaustion processes are again exhaustion processes under their corresponding exhaustion semantics.

5 Finite Exhaustion Theorems

5.1 The Reachable Evolving Graph

Let

$$\mathrm{Reach}(q_0) \subseteq Q$$

denote the set of states reachable from q_0. Define the reachable evolving graph

$$G_{\mathrm{ev}}(\mathcal{E})$$

as the directed graph induced by

$$\mathrm{Reach}(q_0) \cap Q^{\mathrm{ev}}.$$

Theorem 5.1 (Finite exhaustion equivalence)**.** *Let $\mathcal{E}$ be a finite exhaustion machine. The following are equivalent:*

1. *Every admissible run from q_0 exhausts in finite time.*
2. *There is no directed cycle in $G_{\mathrm{ev}}(\mathcal{E})$.*
3. *Every admissible run from q_0 has length at most $|\mathrm{Reach}(q_0) \cap Q^{\mathrm{ev}}|$ before exhaustion.*
4. *There exists a rank function*

$$R : \mathrm{Reach}(q_0) \cap Q^{\mathrm{ev}} \to \mathbb{N}$$

such that

$$q \to q', \quad q, q' \in Q^{\mathrm{ev}} \quad \Longrightarrow \quad R(q') < R(q).$$

Proof. (1) ⇒ (2). Suppose $G_{\mathrm{ev}}(\mathcal{E})$ contains a reachable directed cycle

$$q_1 \to q_2 \to \cdots \to q_k \to q_1$$

with all $q_i \in Q^{\mathrm{ev}}$. Since the cycle is reachable, some admissible path reaches q_1. After that, one may repeatedly follow the labels realizing the cycle. This gives an infinite admissible run that never enters Q^{ex}, contradicting universal exhaustion.

(2) ⇒ (3). If $G_{\mathrm{ev}}(\mathcal{E})$ is acyclic, no path inside it can visit the same evolving state twice. Therefore every path through evolving states has length at most the number of reachable evolving states. Hence every admissible run exhausts or becomes undefined within at most $|\mathrm{Reach}(q_0) \cap Q^{\mathrm{ev}}|$ evolving transitions.

(3) ⇒ (1). Immediate, since a uniform finite bound on run length implies finite exhaustion.

(2) ⇒ (4). In a finite directed acyclic graph, define $R(q)$ to be the maximum length of a directed path beginning at q and remaining inside $G_{\mathrm{ev}}(\mathcal{E})$. If $q \to q'$, then every path from q' extends to a path from q by adding the edge $q \to q'$. Hence

$$R(q) \geq R(q') + 1,$$

so $R(q') < R(q)$.

(4) ⇒ (2). If a directed cycle existed,

$$q_1 \to q_2 \to \cdots \to q_k \to q_1,$$

then the rank condition would imply

$$R(q_2) < R(q_1), \quad R(q_3) < R(q_2), \quad \ldots, \quad R(q_1) < R(q_k).$$

Combining these gives

$$R(q_1) < R(q_1),$$

a contradiction. Therefore no directed cycle exists. □

Corollary 5.2 (Uniform exhaustion bound). *If $\mathcal{E}$ is universally exhausting from q_0, then*

$$\tau \leq |\operatorname{Reach}(q_0) \cap Q^{\mathrm{ev}}|$$

for every admissible run.

Proof. This is item 3 of the finite exhaustion equivalence theorem. □

5.2 Well-Founded Ranking

Theorem 5.3 (Well-founded exhaustion theorem). *Let W be a well-founded ordered set. Suppose there exists a map*

$$R : Q^{\mathrm{ev}} \to W$$

such that every evolving transition strictly decreases R:

$$q \to q', \quad q, q' \in Q^{\mathrm{ev}} \quad \Longrightarrow \quad R(q') < R(q).$$

Then every admissible run exhausts in finite time.

Proof. Assume for contradiction that there exists an infinite admissible run

$$q_0 \to q_1 \to q_2 \to \cdots$$

remaining forever inside Q^{ev}. Then

$$R(q_0) > R(q_1) > R(q_2) > \cdots$$

is an infinite descending chain in W, contradicting well-foundedness. Hence no infinite evolving run exists. Therefore every admissible run exhausts after finitely many steps. □

Corollary 5.4 (Resource-vector exhaustion). *Let*

$$r : Q \to \mathbb{N}^k$$

be a resource vector ordered lexicographically. If every evolving transition satisfies

$$r(q') <_{\mathrm{lex}} r(q),$$

then the machine exhausts in finite time.

Proof. The lexicographic order on $\mathbb{N}^k$ is well-founded. Apply the well-founded exhaustion theorem with $R = r$. □

6 The Halting Problem Inside Exhaustion Theory

The classical Halting Problem is undecidable for Turing machines because Turing machines have unbounded computational configurations. Exhaustion machines are different: their evolving region is finite and consumable.

Definition 6.1 (Halting problem for exhaustion machines). Given a finite exhaustion machine $\mathcal{E}$, decide whether every admissible run from q_0 reaches Q^{ex} or becomes undefined after finitely many steps.

Theorem 6.2 (Decidability of exhaustion halting). *The Halting Problem for finite exhaustion machines is decidable.*

Proof. Construct the reachable evolving graph $G_{\mathrm{ev}}(\mathcal{E})$. By the finite exhaustion equivalence theorem, $\mathcal{E}$ halts on every admissible run if and only if $G_{\mathrm{ev}}(\mathcal{E})$ contains no directed cycle. Since $G_{\mathrm{ev}}(\mathcal{E})$ is finite, cycle detection is decidable by depth-first search or topological sorting. Therefore the halting problem for finite exhaustion machines is decidable. $\square$

Corollary 6.3 (Graph-theoretic halting)**.** *For genuine finite exhaustion machines, halting is not an undecidable computation-theoretic property. It is the graph-theoretic property*

$$\boxed{\textit{no reachable cycle inside } Q^{\mathrm{ev}}.}$$

Proof. Immediate from the preceding theorem. $\square$

7 Exhaustion Profiles and Frontiers

7.1 Remaining Life

Definition 7.1 (Remaining life)**.** For $q \in Q$, the remaining life $L(q)$ is the supremum of the lengths of admissible paths beginning at q before exhaustion:

$$L(q) = \sup\{m : q = q_0 \to q_1 \to \cdots \to q_m,\ q_i \in Q^{\mathrm{ev}} \text{ for } 0 \le i < m\}.$$

If $q \in Q^{\mathrm{ex}}$, set

$$L(q) = 0.$$

In a universally exhausting finite machine, $L(q) < \infty$ for every reachable state.

Definition 7.2 (Exhaustion profile)**.** The exhaustion profile of $\mathcal{E}$ is

$$\mathrm{Prof}(\mathcal{E}) = \{L(q) : q \in \mathrm{Reach}(q_0)\}.$$

Proposition 7.3 (Remaining life is a canonical rank)**.** *If $\mathcal{E}$ is universally exhausting, then for every evolving transition*

$$q \to q'$$

with $q, q' \in Q^{\mathrm{ev}}$,

$$L(q') < L(q).$$

Proof. Every admissible path from q' may be extended by the edge $q \to q'$ to form an admissible path from q. Thus

$$L(q) \ge 1 + L(q').$$

Therefore $L(q') < L(q)$. $\square$

7.2 The Exhaustion Frontier

Definition 7.4 (Exhaustion frontier)**.** The exhaustion frontier is

$$\partial_{\mathrm{ex}} Q = \{q \in Q^{\mathrm{ev}} : \exists a \in \Sigma \text{ such that } \delta(q,a) \in Q^{\mathrm{ex}} \text{ or } \delta(q,a) \text{ is undefined}\}.$$

The frontier consists of evolving states one transition away from exhaustion or undefined continuation.

Proposition 7.5 (Frontier intersection)**.** *Every nontrivial finite run that exhausts must pass through the exhaustion frontier immediately before exhaustion.*

Proof. Let

$$q_0 \to q_1 \to \cdots \to q_T$$

be a run with $q_T \in Q^{\mathrm{ex}}$ or no admissible continuation at q_T, and suppose $T > 0$. Then $q_{T-1} \in Q^{\mathrm{ev}}$, and the final transition either enters Q^{ex} or leads to a state with no admissible continuation. Hence $q_{T-1} \in \partial_{\mathrm{ex}} Q$. $\square$

8 Algebra of Exhaustion

8.1 Sequential Composition

Let $\mathcal{E}$ and $\mathcal{F}$ be deterministic exhaustion processes with deterministic exhaustion times $\tau_{\mathcal{E}}$ and $\tau_{\mathcal{F}}$. Their sequential composition

$$\mathcal{E};\mathcal{F}$$

means: run $\mathcal{E}$ until exhaustion, then initialize $\mathcal{F}$.

Proposition 8.1 (Additivity of sequential exhaustion)**.** *For deterministic finite-horizon processes,*

$$\tau_{\mathcal{E};\mathcal{F}} = \tau_{\mathcal{E}} + \tau_{\mathcal{F}}.$$

Proof. By definition, the composed process first spends exactly $\tau_{\mathcal{E}}$ steps in $\mathcal{E}$, then exactly $\tau_{\mathcal{F}}$ steps in $\mathcal{F}$. Therefore its total duration is the sum. □

8.2 Choice Composition

The choice composition

$$\mathcal{E} \oplus \mathcal{F}$$

means that one of the two processes is selected.

Definition 8.2 (Pessimistic and optimistic choice horizons)**.** For deterministic horizons,

$$\tau_{\min}(\mathcal{E} \oplus \mathcal{F}) = \min\{\tau_{\mathcal{E}}, \tau_{\mathcal{F}}\},$$

and

$$\tau_{\max}(\mathcal{E} \oplus \mathcal{F}) = \max\{\tau_{\mathcal{E}}, \tau_{\mathcal{F}}\}.$$

The pessimistic horizon corresponds to adversarial or worst-case choice. The optimistic horizon corresponds to best-case choice.

8.3 Parallel Composition

The parallel composition

$$\mathcal{E} \otimes \mathcal{F}$$

runs two exhaustion processes together.

Definition 8.3 (Any-exhaustion semantics)**.** Under any-exhaustion semantics, the composite exhausts when either component exhausts:

$$\tau_{\exists}(\mathcal{E} \otimes \mathcal{F}) = \min\{\tau_{\mathcal{E}}, \tau_{\mathcal{F}}\}.$$

Definition 8.4 (All-exhaustion semantics)**.** Under all-exhaustion semantics, the composite exhausts only when both components exhaust:

$$\tau_{\forall}(\mathcal{E} \otimes \mathcal{F}) = \max\{\tau_{\mathcal{E}}, \tau_{\mathcal{F}}\}.$$

Proposition 8.5 (Parallel horizon law)**.** *For deterministic finite-horizon processes,*

$$\tau_{\exists}(\mathcal{E} \otimes \mathcal{F}) = \min\{\tau_{\mathcal{E}}, \tau_{\mathcal{F}}\}, \qquad \tau_{\forall}(\mathcal{E} \otimes \mathcal{F}) = \max\{\tau_{\mathcal{E}}, \tau_{\mathcal{F}}\}.$$

Proof. Under any-exhaustion semantics, the first component to exhaust terminates the composite, so the horizon is the smaller lifetime. Under all-exhaustion semantics, the composite remains available until both components are exhausted, so the horizon is the larger lifetime. □

9 Exhaustion Semirings

9.1 Pessimistic Min-Plus Algebra

Let

$$\mathbb{N}_\infty = \mathbb{N} \cup \{\infty\}.$$

Define

$$a \oplus b = \min\{a, b\}, \qquad a \otimes b = a + b,$$

with the conventions

$$a + \infty = \infty + a = \infty.$$

Theorem 9.1 (Pessimistic exhaustion semiring)**.** *The structure*

$$(\mathbb{N}_\infty, \min, +)$$

is an idempotent semiring with additive identity ∞ *and multiplicative identity* 0*.*

Proof. The operation min is associative, commutative, and idempotent:

$$\min(a, a) = a.$$

Its identity element is ∞, since $\min(a, \infty) = a$. Addition is associative and has identity 0. Addition distributes over minimum:

$$a + \min(b, c) = \min(a + b, a + c),$$

and similarly on the right. Finally, ∞ is absorbing for addition:

$$a + \infty = \infty.$$

Thus $(\mathbb{N}_\infty, \min, +)$ is an idempotent semiring. □

9.2 Optimistic Max-Plus Algebra

To obtain a strict max-plus semiring, include a formal impossible horizon $-\infty$. Let

$$\mathbb{N}_{\pm\infty} = \mathbb{N} \cup \{-\infty, \infty\}.$$

Define

$$a \oplus b = \max\{a, b\}, \qquad a \otimes b = a + b,$$

with

$$-\infty + a = -\infty, \qquad \infty + a = \infty \quad \text{for } a \neq -\infty.$$

Theorem 9.2 (Optimistic exhaustion semiring)**.** *The structure*

$$(\mathbb{N}_{\pm\infty}, \max, +)$$

is an idempotent semiring with additive identity $-\infty$ *and multiplicative identity* 0*, after excluding the indeterminate expression* $\infty + (-\infty)$*.*

Proof. The operation max is associative, commutative, and idempotent. Its identity is $-\infty$, since

$$\max(a, -\infty) = a.$$

Addition is associative on the defined extended-natural expressions and has identity 0. Addition distributes over maximum:

$$a + \max(b, c) = \max(a + b, a + c).$$

Finally, $-\infty$ is absorbing for addition:

$$a + (-\infty) = -\infty.$$

Therefore the max-plus exhaustion algebra is an idempotent semiring on the defined domain. $\square$

Remark 9.3. The pessimistic semiring models adversarial or fragile exhaustion. The optimistic semiring models redundancy, best-case survival, or controlled delay of exhaustion.

10 Morphisms and the Category of Exhaustion Machines

10.1 Exhaustion-Preserving Maps

Definition 10.1 (Exhaustion morphism)**.** Let

$$\mathcal{E} = (Q, \Sigma, \delta, q_0, Q^{\text{ev}}, Q^{\text{ex}})$$

and

$$\mathcal{F} = (P, \Sigma, \gamma, p_0, P^{\text{ev}}, P^{\text{ex}})$$

be exhaustion machines over the same alphabet. An exhaustion morphism

$$f : \mathcal{E} \to \mathcal{F}$$

is a function

$$f : Q \to P$$

such that:

1. $f(q_0) = p_0$;
2. $q \in Q^{\text{ev}} \Rightarrow f(q) \in P^{\text{ev}}$;
3. $q \in Q^{\text{ex}} \Rightarrow f(q) \in P^{\text{ex}}$;
4. whenever $\delta(q, a)$ is defined,
$$f(\delta(q, a)) = \gamma(f(q), a).$$

Theorem 10.2 (Category of exhaustion machines)**.** *Exhaustion machines over a fixed alphabet Σ, together with exhaustion morphisms, form a category* $\mathbf{Exh}_\Sigma$.

Proof. The identity map

$$\mathrm{id}_Q : Q \to Q$$

preserves the initial state, evolving states, exhausted states, and transitions. Hence it is an exhaustion morphism.

Let

$$f : \mathcal{E} \to \mathcal{F} \qquad \text{and} \qquad g : \mathcal{F} \to \mathcal{G}$$

be exhaustion morphisms. Their composition $g \circ f$ preserves the initial state:

$$(g \circ f)(q_0) = g(p_0) = r_0.$$

It preserves evolving states because f sends evolving states to evolving states and g does the same. It preserves exhausted states by the same reasoning. Finally, if $\delta(q, a)$ is defined, then

$$(g \circ f)(\delta(q, a)) = g(f(\delta(q, a))) = g(\gamma(f(q), a)) = \eta(g(f(q)), a),$$

where η is the transition map of $\mathcal{G}$. Therefore $g \circ f$ is an exhaustion morphism. Associativity follows from associativity of function composition. Hence $\mathbf{Exh}_\Sigma$ is a category. $\square$

10.2 Collapse Skeleton

Definition 10.3 (Collapse map)**.** The collapse map of an exhaustion machine is

$$\kappa : Q \to \{E, X\}$$

defined by

$$\kappa(q) = \begin{cases} E, & q \in Q^{\mathrm{ev}}, \\ X, & q \in Q^{\mathrm{ex}}. \end{cases}$$

The two-state skeleton is:

$$E \longrightarrow X.$$

Proposition 10.4 (Universal exhaustion abstraction)**.** *Every exhaustion machine admits a unique partition-preserving map to the two-state exhaustion skeleton.*

Proof. Since $Q = Q^{\mathrm{ev}} \sqcup Q^{\mathrm{ex}}$, every $q \in Q$ is either evolving or exhausted, but not both. Therefore the map κ is uniquely determined by the requirement that evolving states map to E and exhausted states map to X. □

11 Algorithms

The algorithms below are written without specialized algorithm packages.

Algorithm 11.1 (Validation of universal exhaustion)**.** Given a finite exhaustion machine $\mathcal{E}$, decide whether every admissible run exhausts.

Step	Operation
1	Construct the directed transition graph $G = (Q, T)$.
2	Compute $\mathrm{Reach}(q_0)$ by breadth-first search or depth-first search.
3	Restrict G to $\mathrm{Reach}(q_0) \cap Q^{\mathrm{ev}}$.
4	Check whether this restricted graph has a directed cycle.
5	If no cycle exists, return "universally exhausting".
6	If a cycle exists, return "not universally exhausting".

The running time is

$$O(|Q| + |T|).$$

Correctness. By the finite exhaustion equivalence theorem, universal exhaustion is equivalent to absence of reachable cycles inside the evolving subgraph. The algorithm checks exactly this condition. Breadth-first search and cycle detection are correct for finite directed graphs. Therefore the algorithm is correct. □

Algorithm 11.2 (Computation of remaining life)**.** Given a universally exhausting finite machine, compute $L(q)$ for every reachable state.

Step	Operation
1	Construct $G_{\mathrm{ev}}(\mathcal{E})$.
2	Topologically sort $G_{\mathrm{ev}}(\mathcal{E})$.
3	Set $L(q) = 0$ for all $q \in Q^{\mathrm{ex}}$.
4	Process evolving states in reverse topological order.
5	For each $q \in Q^{\mathrm{ev}}$, set $L(q) = 1 + \max\{L(q') : q \to q'\}$.
6	If q has no outgoing admissible transition, set $L(q) = 0$.

Correctness. Since the machine is universally exhausting, the reachable evolving graph is acyclic. Reverse topological order ensures that $L(q')$ has already been computed for every successor q' of q. The recurrence

$$L(q) = 1 + \max_{q \to q'} L(q')$$

is exactly the definition of the longest admissible remaining path. Thus the algorithm computes the remaining life of every reachable state. □

Algorithm 11.3 (Maximum-delay policy)**.** Given a controlled exhaustion machine, choose actions that delay exhaustion maximally.

Step	Operation
1	Compute $L(q)$ for all reachable states.
2	At state q, inspect all admissible actions a.
3	Choose $a^* \in \arg\max_a L(\delta(q,a))$.
4	Apply a^* and repeat until exhaustion.

Correctness. The function $L(q)$ gives the maximum possible remaining life from q. Choosing an action leading to a successor with maximal L realizes the dynamic-programming recurrence for $L(q)$. Therefore the policy delays exhaustion maximally among deterministic state-feedback policies. □

12 Vector Graphics

12.1 The Primitive Exhaustion Skeleton

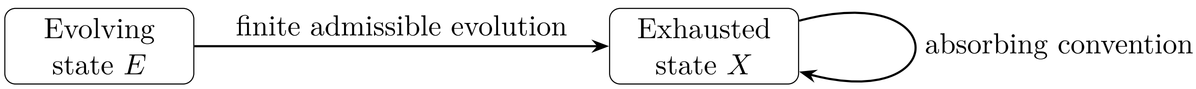

Figure 1: The primitive skeleton of Exhaustion Theory. Under strict terminal semantics, the self-loop at X is removed.

12.2 Acyclic Evolving Region and Exhaustion Frontier

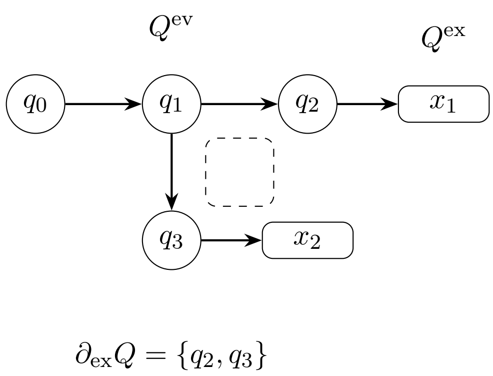

Figure 2: A finite acyclic evolving region. The frontier consists of evolving states one step from exhaustion.

12.3 Sequential and Parallel Exhaustion

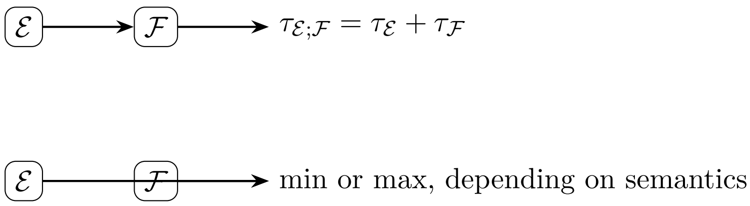

Figure 3: Sequential composition gives additive lifetime; parallel composition gives minimum or maximum lifetime depending on fragility or redundancy.

12.4 Joint Ambiguity–Exhaustion Diagram

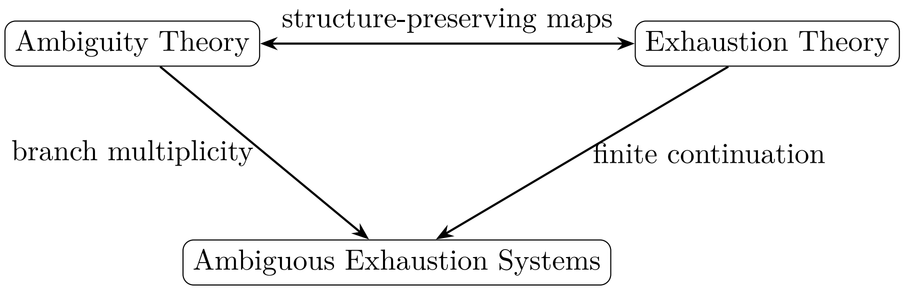

Figure 4: Ambiguity Theory studies multiplicity across alternatives; Exhaustion Theory studies finite continuation along time. Their joint theory studies branchwise finite processes.

13 Applications

13.1 Resource-Bounded Computation

A resource-bounded computation has a finite budget:

$$r(q) \in \mathbb{N}^k.$$

The computation is exhausting if every meaningful transition decreases this budget:

$$r(q') <_{\text{lex}} r(q).$$

Examples include bounded-memory programs, finite-stack parsers, bounded proof search, bounded model checking, and finite-token inference.

The exhausted state is not necessarily accept or reject. It may be:

out of memory, out of time, out of stack, out of admissible hypotheses.

13.2 Thermodynamic Exhaustion

Let

$$F(q)$$

be the available free energy or usable gradient in a coarse-grained thermodynamic state. A dissipative process is exhausting when admissible transitions satisfy

$$F(q') < F(q).$$

The exhausted state is the state in which no usable work can be extracted:

$$F(q) = 0.$$

This is not merely equilibrium. Equilibrium means no net macroscopic change. Exhaustion means no admissible productive continuation relative to a chosen resource.

13.3 Finite-Horizon Control

In finite-horizon control, a controller seeks to delay, accelerate, or shape exhaustion. The maximum-delay value function is exactly the remaining life:

$$L(q) = 1 + \max_a L(\delta(q, a)).$$

The minimum-time-to-exhaustion value function is

$$M(q) = 1 + \min_a M(\delta(q, a)).$$

Thus Exhaustion Theory gives a natural bridge between automata, dynamic programming, and controlled resource depletion.

13.4 AI and Machine Learning

In machine learning, exhaustion appears whenever a finite admissible class is progressively depleted.

- In version-space learning, hypotheses inconsistent with data are removed until the version space exhausts or collapses to a singleton.
- In decision-tree learning, splitting continues until impurity, depth budget, or sample support is exhausted.
- In reinforcement learning, an agent may operate under finite energy, finite memory, finite exploration budget, or finite safe-action budget.
- In large language model inference, generation is bounded by a token budget; continuation exhausts when the budget or admissible context is depleted.
- In neural architecture search, a finite search budget induces an exhaustion machine over candidate architectures.

AI/ML setting	Evolving state	Exhausted state	Rank/resource
Version space	Consistent hypotheses remain	No hypothesis or singleton	$\lvert V(D)\rvert$
Decision tree	Splits available	No valid split	Depth/sample budget
Reinforcement learning	Safe actions remain	No safe action	Energy/time budget
LLM inference	Tokens remain	Context or token budget depleted	Remaining tokens
Search/planning	Frontier nonempty	Frontier empty	Open nodes

Table 1: Examples of exhaustion structure in artificial intelligence and machine learning.

14 Structural Summary Tables

Concept	Exhaustion Theory	Interpretation
Primitive object	Exhaustion machine	Finite process with terminal regime
State classes	$Q^{\mathrm{ev}}, Q^{\mathrm{ex}}$	Evolving and exhausted states
Main invariant	$L(q)$	Remaining admissible life
Main obstruction	Reachable cycle in Q^{ev}	Non-exhausting continuation
Algebra	$\min, +$ or $\max, +$	Pessimistic or optimistic horizon
Morphisms	Exhaustion-preserving maps	Preserve dynamics and terminality
Decision problem	Cycle detection	Decidable halting

Table 2: Basic dictionary of Exhaustion Theory.

Framework	What fails?	Exhaustion-theoretic distinction
Limit theory	Infinite-time convergence	Exhaustion is finite-time termination
Singularity theory	Regularity or boundedness	Exhaustion need not blow up
Automata theory	Acceptance/rejection	Exhaustion is loss of continuation
Computability theory	Halting undecidability	Exhaustion halting is decidable
Thermodynamics	Free energy depletion	Exhaustion tracks usable continuation
Reliability theory	Component failure	Exhaustion generalizes finite depletion

Table 3: Comparison with neighbouring mathematical frameworks.

15 Joint Framework with Ambiguity Theory

Ambiguity Theory and Exhaustion Theory are orthogonal but compatible.

Ambiguity Theory studies objects of the form

$$A = \langle a_1 \mid a_2 \mid \cdots \mid a_n \rangle,$$

where no branch is internally selected. Exhaustion Theory studies processes

$$q_0 \to q_1 \to \cdots \to q_T \in Q^{\mathrm{ex}}$$

whose admissible continuation terminates.

Thus:

$$\boxed{\text{Ambiguity is multiplicity across branches.}}$$

$$\boxed{\text{Exhaustion is finitude along time.}}$$

15.1 Ambiguous Exhaustion States

Definition 15.1 (Ambiguous exhaustion state)**.** Let $\mathcal{E}$ be an exhaustion machine. An ambiguous exhaustion state is an ambi-object over Q:

$$A = \langle q_1 \mid q_2 \mid \cdots \mid q_n \rangle.$$

The branches may have different exhaustion statuses. Define:

$$A \in \mathbf{Amb}(Q^{\mathrm{ev}})$$

if every branch lies in Q^{ev}, and

$$A \in \mathbf{Amb}(Q^{\mathrm{ex}})$$

if every branch lies in Q^{ex}.

A mixed object such as

$$\langle q_{\mathrm{ev}} \mid q_{\mathrm{ex}} \rangle$$

is structurally ambiguous over exhaustion status itself.

15.2 Branchwise Exhaustion Update

Given an action $a \in \Sigma$, define the branchwise update

$$\widehat{\delta}(A, a) = \langle \delta(q_1, a) \mid \cdots \mid \delta(q_n, a) \rangle,$$

where undefined branches are sent to a formal exhausted symbol $X_\perp$.

This is the exhaustion analogue of ambiguity-preserving dynamics.

Theorem 15.2 (Branchwise exhaustion theorem)**.** *Let*

$$A = \langle q_1 \mid \cdots \mid q_n \rangle$$

be an ambiguous exhaustion state, and suppose each q_i belongs to a universally exhausting finite machine. Then under branchwise evolution:

$$\tau_\forall(A) = \max_i L(q_i), \qquad \tau_\exists(A) = \min_i L(q_i).$$

Proof. Under all-branches semantics, the ambiguous process is fully exhausted only when every branch has exhausted. Since branch i lasts $L(q_i)$ steps, the time at which all branches have exhausted is the maximum of the branch lifetimes:

$$\tau_\forall(A) = \max_i L(q_i).$$

Under some-branch semantics, the ambiguous process has at least one exhausted branch as soon as the shortest-lived branch exhausts. Hence

$$\tau_\exists(A) = \min_i L(q_i).$$

□

15.3 Ambiguity–Exhaustion Table

Aspect	Ambiguity Theory	Exhaustion Theory
Primitive	Ambi-object $\langle a_1 \vert \cdots \vert a_n \rangle$	Exhaustion machine $\mathcal{E}$
Core distinction	Branch multiplicity	Evolving/exhausted partition
Main operation	Resolution R_i	Termination/exhaustion τ
Information loss	Selecting one branch	Losing future continuation
Algebra	Merge, nesting, coupling	Sequential, choice, parallel
Invariant	Ambiguity profile	Exhaustion profile
Graph object	Coupling graph	Evolving transition graph
Failure mode	Premature resolution	Reachable evolving cycle

Table 4: Ambiguity Theory and Exhaustion Theory as companion frameworks.

15.4 Joint Research Principle

The joint theory studies objects of the form

$$\mathcal{J} = (\mathcal{E}, A),$$

where $\mathcal{E}$ is an exhaustion machine and A is an ambi-object over its state space. The central question becomes:

$$\boxed{\text{How does unresolved multiplicity evolve under finite admissible continuation?}}$$

This leads to new invariants:

$$\mathrm{Prof}_{\mathbf{Amb},\mathbf{Exh}}(A) = \langle L(q_1) \mid \cdots \mid L(q_n) \rangle,$$

the ambiguous lifetime profile.

16 Open Problems

P1. Classification. Classify finite exhaustion machines up to exhaustion-preserving isomorphism. Determine whether the exhaustion profile and transition partial order form a complete invariant.

P2. Minimal rank. Given a universally exhausting machine, find the minimal well-founded rank representing its exhaustion structure.

P3. Algebraic completeness. Develop a complete algebra for sequential, choice, parallel, and coupled exhaustion.

P4. Coupled exhaustion. Study systems whose components do not exhaust independently because they share a finite resource.

P5. Probabilistic exhaustion. Extend the theory to Markov chains with absorbing exhausted states and study expected exhaustion time.

P6. Ambiguous exhaustion. Develop the full joint theory of ambi-objects evolving over exhaustion machines.

P7. Thermodynamic semantics. Give a physically rigorous account of finite-state thermodynamic exhaustion in terms of free energy, entropy production, and coarse-grained admissible transitions.

P8. Complexity. Classify the computational complexity of optimal exhaustion delay, accelerated exhaustion, and coupled-resource exhaustion.

P9. Category theory. Determine categorical limits, colimits, monoidal structures, and possible internal logics of **Exh**.

P10. Exhaustion learning theory. Develop learning-theoretic bounds in which generalization depends on exhaustion of a finite hypothesis, sample, or computation budget.

17 Conclusion

Exhaustion Theory studies systems that run out. Its primitive mathematical representation is a finite-state process partitioned into evolving states and exhausted states. The central structural theorem is simple and powerful:

$$\boxed{\text{finite universal exhaustion} \iff \text{no reachable cycle inside the evolving subgraph.}}$$

Equivalently, a finite exhaustion process is one that admits a decreasing well-founded rank. This makes the Halting Problem decidable within the class of exhaustion machines: halting is reduced to cycle detection.

The algebra of exhaustion is governed by addition, minimum, and maximum. Sequential composition adds lifetimes. Fragile parallelism and adversarial choice take minimums. Redundant parallelism and optimistic choice take maximums. These operations naturally lead to min-plus and max-plus semiring structures.

The theory has direct interpretations in computation, thermodynamics, artificial intelligence, machine learning, reliability, and finite-horizon control. In its joint form with Ambiguity Theory, it studies branchwise finite continuation: ambiguous systems whose possible states may exhaust at different times.

The companion slogan is therefore:

$$\boxed{\text{Ambiguity Theory studies unresolved alternatives;}}$$

$$\boxed{\text{Exhaustion Theory studies finite futures.}}$$

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Glossary

Absorbing terminal semantics
A representation convention in which an exhausted state has only a self-loop.

Admissible continuation
A transition allowed by the partial transition map of an exhaustion machine.

Ambiguous exhaustion state
An ambi-object whose branches are states of an exhaustion machine.

Any-exhaustion semantics
Parallel semantics in which a composite system exhausts when at least one component exhausts.

Collapse map
The map $\kappa : Q \to \{E, X\}$ sending evolving states to E and exhausted states to X.

Coupled exhaustion
Exhaustion in which two or more subsystems share a finite resource, so their lifetimes are not independent.

Evolving state
A state with admissible endogenous continuation.

Exhausted state
A state with no productive endogenous continuation.

Exhaustion frontier
The set of evolving states one admissible transition away from exhaustion or undefined continuation.

Exhaustion machine
A finite-state process whose state space is partitioned into evolving and exhausted states.

Exhaustion morphism
A map between exhaustion machines preserving initial states, evolving states, exhausted states, and transitions.

Exhaustion profile
The collection of remaining lives $L(q)$ over reachable states.

Exhaustion time
The first time at which a run enters an exhausted state or loses admissible continuation.

Finite admissible continuation
The defining phenomenon of Exhaustion Theory: dynamics are valid for only finitely many steps.

Max-plus exhaustion algebra
The algebra $(\mathbb{N}_{\pm\infty}, \max, +)$, used for optimistic or redundant exhaustion.

Min-plus exhaustion algebra
The algebra $(\mathbb{N}_{\infty}, \min, +)$, used for pessimistic or adversarial exhaustion.

Remaining life
The maximum number of admissible transitions available from a state before exhaustion.

Resource rank
A well-founded quantity that strictly decreases along evolving transitions.

Strict terminal semantics
A representation convention in which exhausted states have no outgoing transitions.

Universal exhaustion
The property that every admissible run from the initial state exhausts in finite time.

Well-founded descent
A ranking condition excluding infinite evolving trajectories.

The End